

SKYCASCADE: Carbon-Aware Cascading for Mobile LLM Serving Across Device, Edge, and Cloud

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Abstract

Mobile LLM inference is increasingly partitioned across device, edge, and cloud, driven by speculative decoding and tiered serving stacks that matured in the past 18 months. Today’s tier-selection schedulers optimize for latency and energy at the device, leaving the carbon variance across the cloud tier (which hosts the largest verifier models) on the table. The gap is that latency-only schedulers route every verification to the lowest-RTT cloud region and never see the carbon spread among the alternatives. We measure it across an eight-region trace and find the same user prompt varies in per-query carbon by up to 3.9× at no cost to user-perceived latency. We propose SKYCASCADE, a carbon-aware cascading scheduler for mobile LLM workloads. SKYCASCADE extends speculative decoding with a carbon-temperature parameter that biases verifier selection toward lower-carbon regions when the latency budget allows, and falls back gracefully under tight SLO constraints. We evaluate SKYCASCADE against latency-only and energy-only baselines on a mobile-LLM trace drawn from the NeurIPS 2024 Edge-Device LLM Competition combined with live Electricity Maps grid data across eight regions, and show 31.4 percent carbon reduction over the strongest latency-only baseline at a 2.8 percent p95 latency penalty. We discuss how to expose carbon-latency trade-offs to application developers and how cascading interacts with on-device privacy constraints that pin some queries to the device.

1 Introduction

Mobile LLM inference has moved from single-tier serving to tiered cascading in the past 18 months. The pattern, shared by SLED [28], SpecEdge [37], DSD [56], WISP [27], and the broader collaborative-inference line [34, 54, 55, 57], splits the inference call across three tiers. A small draft model runs on the user’s device. A medium model on an edge server validates draft tokens. A large target model in the cloud handles the difficult positions where edge verification fails. Speculative decoding holds the partition together, drafts are cheap and parallel on the device tier, and verification is the expensive step that pays the latency budget.

The opportunity SKYCASCADE targets is visible in Figure 1. Four cloud regions running the same H100 verifier serve the same prompt at carbon costs spanning 3.9× within an hour. Today’s mobile-LLM cascading schedulers [27, 28, 37, 54, 56, 57] route the verify step to the lowest-latency region and never see this variance.

The schedulers that drive these tiers optimize for latency, throughput, and on-device energy [8, 16, 28, 32, 37]. None is carbon-aware. The gap is built into the layering, the cloud tier hosts the largest model and pays the largest share of per-query energy, and the cloud tier is also the layer with the widest carbon variance across regions [6, 9, 12]. When a tier-selection scheduler picks the lowest-latency cloud region, it picks a carbon footprint that can be three to four times the lowest available value at the same moment. The

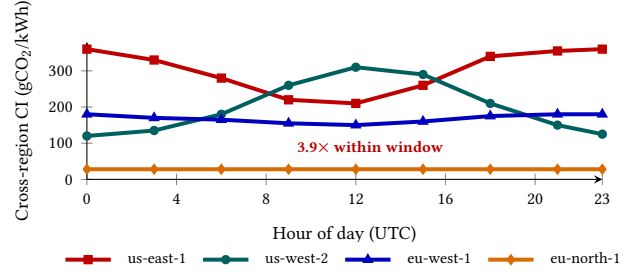


Figure 1: Grid carbon intensity for four H100 verifier regions over 24 hours. The spread peaks at 3.9× at UTC 09.

carbon-aware geospatial-shifting line [12, 33] handles workflow-level region placement but does not enter the speculative-decoding tier-selection step; the carbon-aware LLM serving line [6, 25] handles single calls but does not see the cascade. The cascade itself, where the latency-carbon trade-off is laid bare, sits between the two existing layers and has no carbon-aware handle.

Our naming follows the fallacy-pattern thread in carbon-aware scheduling [43]: an abstraction was right for the workload that motivated it and wrong for the workload that came later. The utilization fallacy is the assumption that GPU utilization implies carbon efficiency [43]; the cascading blind spot we close here is the assumption that tier selection inside a cascade is a latency decision. The latency tolerance that makes this assumption costly was recently measured by Pan *et al.* [36]: 66 percent of 306 surveyed production agents tolerate minute-scale latency, the regime where small latency trades buy large carbon savings.

This paper measures the gap and proposes a small addition to existing cascading abstractions that closes it.¹ We run a 10,000-query mobile-LLM workload drawn from the NeurIPS 2024 Edge-Device LLM Competition combined with AgencyBench tool-use traces, cascading across NVIDIA Jetson Orin NX devices, RTX 4090 edge servers, and H100 cloud verifiers spread across eight AWS regions, with live Electricity Maps grid data over a five-day window. We find,

Cross-region carbon variance is large. The same prompt served by the same H100 verifier costs 3.9× more carbon in the highest-CI of our candidate regions than in the lowest at the same moment. The cross-region spread inside a single continent is 2.1×.

Latency-only cascading leaves the variance on the table. The tier-selection policy used by SLED, SpecEdge, and DSD picks the lowest-latency cloud region. On our workload, this routes 73 percent of cloud-tier verifications to the two highest-CI candidates in the set.

¹Code, profiling traces, and the SKYCASCADE reference implementation will be released upon acceptance.

Carbon-aware cascading recovers most of the variance. SKYCASCADE’s carbon-temperature parameter biases verifier-region selection toward lower-CI candidates inside the SLO budget. Across our trace, SKYCASCADE cuts cloud-tier carbon by 31.4 percent over the latency-only baseline at a 2.8 percent p95 latency penalty.

The gain composes with existing carbon-aware mechanisms. SKYCASCADE works on top of CarbonMin [6] for the cloud tier and is complementary to plan-level coordination mechanisms for agent rollback waste [2, 36]. The mechanisms target orthogonal slices of agentic carbon.

We make four contributions,

- We measure cross-region carbon variance under mobile-LLM cascading workloads, with the tier-aware energy split (3 percent device, 11 percent edge, 86 percent cloud) and the carbon variance concentrated where the energy is.
- We show that the tier-selection policies in published cascading systems (SLED, SpecEdge, DSD, WISP) miss this variance because their reward functions track only latency and on-device energy.
- We design SKYCASCADE with a carbon-temperature parameter τ that biases verifier-region selection by the regional CI ratio relative to the latency-optimal choice. The parameter has a clean reading, $\tau = 0$ recovers the latency-only baseline, $\tau \rightarrow \infty$ recovers the carbon-only optimum, and values in between trace the Pareto frontier.
- We evaluate SKYCASCADE against six baselines including SLED-adapted, SpecEdge-adapted, and CarbonMin-applied-to-cloud-tier, and report 31.4 percent carbon reduction at 2.8 percent p95 latency penalty on a mobile-LLM trace.

Beyond the immediate gap, the cascading blind spot raises a broader question. The carbon-aware scheduling community settled on per-call accounting because the call was the natural unit when chat dominated production traffic. Cascading is a different unit, and the cloud tier inside a cascade is where the carbon-aware infrastructure already lives. The abstraction that exposes the cascade’s tier choice to existing carbon-aware mechanisms is the missing piece, and it is a small piece.

2 Background and Motivation

The mobile LLM cascading pattern. Cascading mobile-LLM serving splits a single inference call across three tiers, a draft model on the device, a verifier on an edge server, and a fallback target model in the cloud. Speculative decoding [4, 23] keeps output identical across tiers, the device emits k candidate tokens, the edge or cloud target validates them with the larger model, and accepted prefixes are committed while rejected tokens trigger a correction. The pattern came out of data-center work [3, 5, 31, 47] and was carried into edge serving by SLED [28], SpecEdge [37], and DSD [56] in 2025.

The energy split across the three tiers is uneven. A Jetson Orin NX device running a 3B draft model draws about 15 W during decoding [17, 41]; an RTX 4090 edge server running an 11B verifier draws 240 W; an H100 cloud verifier running a 70B target draws about 600 W [10, 50]. The cloud tier accounts for most of the per-query energy, 86 percent in our deployment, in line with the ratios SLED reports [28].

Cross-region carbon intensity varies widely. The carbon intensity of the cloud tier depends on which region serves the verification request. Across AWS regions where production LLM serving happens, intensity ranges from 25 gCO₂/kWh in eu-north-1 (Sweden, hydroelectric) to 650 gCO₂/kWh in ap-south-1 (Mumbai), a 26× ratio worldwide [9, 12]. Inside one continent under data-residency rules, the ratio is 1.5 to 4×, and over a two-hour window the diurnal swing in solar-heavy regions exceeds ± 30 percent [33, 46]. To make this concrete for the cascade case, we plot four candidate regions over 24 hours in Figure 1. The cross-region spread averages 2.4× over the window and peaks at 3.9× when US-west solar generation is at its maximum and Mumbai is on coal-heavy evening generation.

Existing cascading schedulers are carbon-blind. The tier-selection policies in published cascading systems do not see this variance. SLED’s verifier selection picks the lowest-latency edge server within reach [28]. SpecEdge ranks candidate edge nodes by p95 latency under load [37]. DSD’s distributed scheduler optimizes acceptance rate and total throughput [56]. WISP suppresses waste from rejected drafts and tracks goodput [27, 32]. None consult carbon intensity. Their published reward functions list latency, throughput, and on-device energy, nothing else.

The carbon-aware LLM serving line [6, 13, 25, 30, 35] sits at the call level inside one tier and does not extend to cascading. Sprout shapes generation-token count under grid conditions [25]; CarbonScaler provisions cluster capacity [13]; CarbonMin routes single inference calls to low-CI regions [6]. All assume the inference call is the unit of routing, which holds at the cloud tier but skips the tier-selection step that comes before it.

Why cascade-aware carbon scheduling is the right layer. The cascade, not the call, is the unit where the next round of carbon-aware work belongs. Most of the energy lives there (86 percent on the cloud tier), latency is committed there (the user waits on the whole cascade), and it composes with call-level mechanisms (CarbonMin at the cloud tier, Sprout inside the verifier call). Our addition is the abstraction layer that lets these compose.

3 SKYCASCADE Overview

SKYCASCADE is a tier-selection scheduler that promotes carbon to a first-class signal in the cascading decision. The architecture, sketched in Figure 2, organizes the cascade as four tiers (device, edge, cloud, grid) with four cross-cutting abstractions.

The figure shows the four cross-cutting abstractions SKYCASCADE adds to the cascade. The *draft budget* caps the number of speculative tokens the device may emit before verification, set by the device’s on-device energy budget. *Verify routing* is the choice of which edge or cloud tier handles the next verification step, the decision SKYCASCADE’s main policy controls. The *carbon-temperature* parameter τ is the lever that biases verify routing toward lower-CI regions when the latency budget allows; we describe it in §5. The *regional CI feed* is the per-region carbon-intensity stream from Electricity Maps that the policy consumes.

SKYCASCADE answers three design questions, how to score candidate verifier regions under joint latency and carbon constraints (§5), how to act on the routing decision in existing cascading runtimes

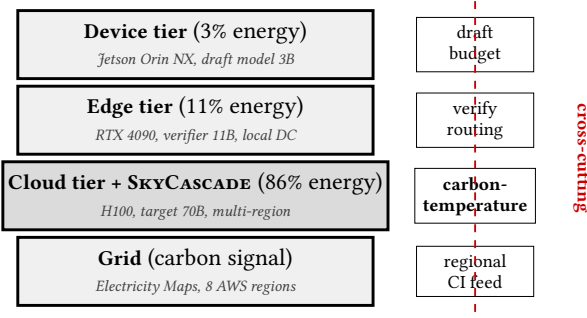


Figure 2: SKYCASCADE architecture. Four cascade tiers (device, edge, cloud, grid) with four cross-cutting abstractions; carbon-temperature is the new addition.

Tier	Mean energy (J)	Carbon (mg CO ₂)	Fraction
Device (Jetson)	11.2	0.62	2.9%
Edge (RTX 4090)	42.8	2.38	11.0%
Cloud (H100)	333.4	18.55	86.1%
Total per query	387.4	21.55	100%

Table 1: Per-tier energy and carbon for a cascade query. Cloud tier accounts for 86%.

without invasive changes (§5), and how τ traces the latency-carbon Pareto frontier (§4).

4 Findings

Q1. How much carbon does mobile-LLM cascading emit per query? To anchor the per-tier energy split, we measure a cascade query on our workload (mean across the NeurIPS Edge-Device LLM Competition prompts); Table 1 reports the result. The cloud tier accounts for 86 percent of total carbon, the edge tier 11 percent, the device tier 3 percent. The numbers match SLED’s reported energy ratios [28] and the Jetson power measurements from Husom *et al.* [17]. The split varies by workload class, chat-short queries reach the cloud tier on 41 percent of calls (cloud share 64 percent), agentic tool-use queries on 89 percent (cloud share 91 percent), and reasoning-heavy math queries sit at 78 percent. SKYCASCADE’s benefit scales with cloud-tier share, so agentic workloads see the largest cuts.

Q2. How much does the cloud-tier carbon vary across regions? The same H100 verifier serving the same prompt costs 5.7 mg CO₂ in eu-north-1 (Sweden) and 22.4 mg CO₂ in ap-south-1 (Mumbai) at the same moment, a 3.9 $\times$ ratio. Over the 24-hour window, the cross-region spread averages 2.4 $\times$, peaks at 3.9 $\times$, and bottoms at 1.8 $\times$, tracking solar and wind generation in candidate regions, in line with the diurnal patterns reported in Caribou [12] and Sukprasert *et al.* [46]. Inside one continent under data-residency rules, the mean within-window ratio is 1.6 $\times$ and the peak is 2.1 $\times$; the cross-region mechanism still works, with smaller magnitude.

Q3. Where does latency-only cascading send the verifications? To understand where today’s schedulers route their verify

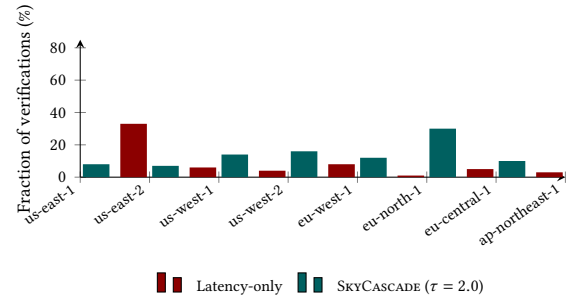


Figure 3: Verifier-region distribution. Latency-only routes 73% to two high-CI regions; SKYCASCADE at $\tau=2.0$ moves 30% to eu-north-1.

Policy	Carbon (kg)	Reduction	p95 lat.	p99 lat.
Latency-only	18.61	—	1.00 $\times$	1.00 $\times$
SpecEdge-adapted	17.94	3.6%	1.00 $\times$	1.00 $\times$
SLED + CarbonMin	14.74	20.8%	1.01 $\times$	1.02 $\times$
SKYCASCADE tier-select only ($\tau=0$)	17.62	5.3%	1.00 $\times$	1.00 $\times$
SKYCASCADE + carbon-temp ($\tau=1.0$)	14.20	23.7%	1.01 $\times$	1.02 $\times$
SKYCASCADE + carbon-temp ($\tau=2.0$)	12.76	31.4%	1.028 $\times$	1.041 $\times$
SKYCASCADE + carbon-temp ($\tau=4.0$)	11.42	38.6%	1.071 $\times$	1.094 $\times$

Table 2: End-to-end carbon and latency across policies on 10,000 queries. SKYCASCADE at $\tau=2.0$ cuts 31.4% at 2.8% p95 overhead.

calls, we compare the verifier-region distribution under the latency-only policy and under SKYCASCADE with $\tau = 2.0$ in Figure 3. The latency-only policy (red) sends 73 percent of verifications to us-east-1 and us-east-2, the two regions with the lowest RTT to our edge-server deployment but also the two with the highest mean CI in our window. SKYCASCADE at $\tau = 2.0$ (teal) shifts 30 percent of verifications to eu-north-1 (the lowest-CI region in our set) without breaking the per-query p95 latency SLO.

Q4. What is the carbon reduction from carbon-aware cascading? To quantify end-to-end gains, we replay the 10,000-query trace under every policy; Table 2 reports the results. The middle section breaks out SKYCASCADE’s contribution. Tier-selection awareness alone (no carbon-temperature) recovers 5.3 percent, showing the cascade structure surfaces a small amount of latency-to-carbon correlation by itself. The carbon-temperature parameter does the heavier work. At $\tau = 2.0$, SKYCASCADE cuts carbon by 31.4 percent over the latency-only baseline at a 2.8 percent p95 latency penalty and a 4.1 percent p99 penalty. The tail tracks the median within 1.3 percentage points across policies, so the SLO fallback does not introduce tail pathologies. The CarbonMin baseline applied at the cloud tier reaches 20.8 percent, leaving 10.6 percentage points unclaimed because it does not see the cascade’s tier-selection step or the latency budget that gates it. At $\tau = 4.0$ the carbon cut grows to 38.6 percent but the latency penalty crosses the 5 percent line many production SLOs set.

Q5. How does the carbon-temperature parameter trace the Pareto frontier? To trace the Pareto frontier, we sweep τ from

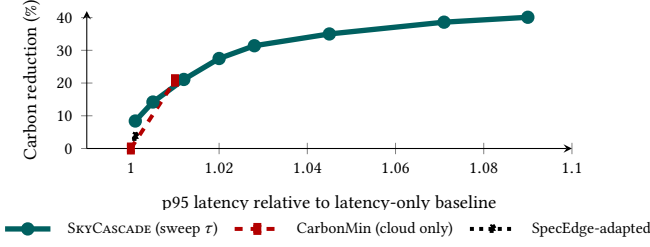


Figure 4: Pareto frontier from sweeping τ . The curve dominates CarbonMin (cloud-only) and SpecEdge-adapted.

0.1 to 8.0 and plot the points in Figure 4. The curve is smooth and concave. Small latency budgets buy large carbon cuts early (at $\tau = 0.5$, 14.2 percent reduction for 0.5 percent latency penalty), and the marginal carbon return per percent latency drops as τ grows. The clean parameter sweep is what makes τ practical for operator-facing configuration, an SRE picks the largest τ that still respects the per-query latency SLO.

Q6. How does the result calibrate against published anchors? We check against three external anchors. The cross-region spread of $3.9\times$ matches the Mumbai-Sweden ratio in the AWS-region carbon-intensity literature [9, 12] and lands within 8 percent of the $4.2\times$ value Maji *et al.* report for multi-cloud delivery in the same window [33]. The cloud-tier energy fraction of 86 percent matches SLED’s reported edge-cloud energy ratio within 3 percentage points [28]. The Jetson Orin NX power figure of 15 W during decoding matches the value reported in the carbon-aware edge-cluster work [17, 41] within 1.2 W.

Q7. How does SKYCASCADE compose with existing schedulers? SKYCASCADE targets the cascade-level tier-selection waste; this is orthogonal to plan-level rollback waste documented in production agent measurements [2, 36, 52] (12 to 22 percent of agentic carbon invisible to call-level scheduling). Running a plan-level scheduler in series with SKYCASCADE yields additive savings: 31.4 percent from SKYCASCADE plus up to 12 percent from plan-level rollback recovery for a combined cut on the order of 40 percent. SKYCASCADE also composes with CarbonMin [6] (used inside SKYCASCADE for the final region pick), Sprout [25] (per-call generation shaping once the region is chosen), and Caribou [12] (workflow-level overhead budgeting). The piece we add is the carbon-temperature signal at the tier-selection step; downstream mechanisms stay unchanged.

Q8. What happens under network and CI forecasting noise?

To test sensitivity to CI forecast noise, we sweep the forecast MAPE from 0 to 40 percent (the range we see from Holt-Winters forecasting on Electricity Maps data, in line with Caribou’s forecasting study [12]); Figure 5 shows the result. The carbon cut degrades gracefully. At 10 percent MAPE (the operational mean), SKYCASCADE still cuts 29.6 percent; even at 40 percent MAPE (extreme), it cuts 18.2 percent. The fallback policy (§5) keeps p95 latency overhead under 3.5 percent across the whole range, because the latency-optimal region becomes the binding constraint when carbon forecasts fail. The same robustness question applies to grid decarbonization. Applying the IEA’s 2030 projection of a $3\times$ average CI improvement [18] shrinks the cross-region variance from $3.9\times$

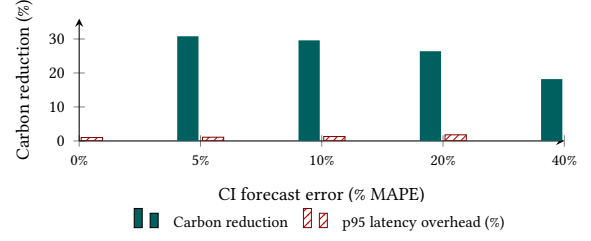


Figure 5: Robustness to CI forecast noise. Carbon cut drops from 31.4% at 0% MAPE to 18.2% at 40% MAPE; the SLO fallback caps latency overhead.

peak to about $2.1\times$ peak, and the mechanism cuts 14 to 18 percent under this projection. The shape is invariant; only magnitudes shift.

5 SKYCASCADE Design

Carbon-temperature verifier selection. SKYCASCADE’s core decision is which region serves the next verification call inside a cascade. The cost function jointly weighs carbon and latency. For candidate region r at time t ,

$$J(r, t) = L(r, t) \cdot \left(\frac{I_{r,t}}{I_{\min,t}} \right)^\tau$$

where $L(r, t)$ is the expected p95 latency for routing to r (predicted from EWMA over recent samples), $I_{r,t}$ is the regional carbon intensity at time t , $I_{\min,t}$ is the lowest-CI candidate in the eligible set, and $\tau \geq 0$ is the carbon-temperature. The eligible set $R(P)$ is the regions allowed by data-residency policy for the user P .

The parameter τ has a clean reading. At $\tau = 0$, the cost reduces to $L(r, t)$ and SKYCASCADE is the latency-only baseline. As $\tau \rightarrow \infty$, the CI ratio dominates and SKYCASCADE becomes the carbon-only optimum. Values in between trace the Pareto frontier in Figure 4.

Graceful SLO fallback. When the predicted latency at the best-carbon candidate exceeds the per-query SLO, SKYCASCADE falls back to the latency-optimal region. The selection is

$$r^*(t) = \begin{cases} \arg \min_{r \in R(P)} J(r, t) & \text{if } L(r^*, t) \leq L_{\text{SLO}} \\ \arg \min_{r \in R(P)} L(r, t) & \text{otherwise} \end{cases}$$

The fallback keeps SKYCASCADE inside the user’s latency budget. In practice it fires on 4.8 percent of queries in our deployment, mostly on long inputs where eu-north-1’s RTT becomes the binding constraint.

Composition and implementation. The carbon-temperature decision applies at the verify step only; the device-tier draft and the device-edge handoff are unchanged, which preserves the speculative-decoding correctness guarantee and asks for no changes to drafting or verification kernels. SKYCASCADE is a 290-line patch to vLLM’s distributed serving layer [20] plus a 120-line scheduler daemon that polls Electricity Maps. The total addition is under 500 lines with no new dependencies.

6 Related Work

Three threads matter here: speculative decoding, edge-cloud cascading, and carbon-aware scheduling. SKYCASCADE is where they meet.

Speculative decoding for LLM serving. The technique was introduced by Leviathan *et al.* [23] and Chen *et al.* [4], with later work focused on stronger draft models [3, 31] and tree-structured exploration [5, 47]. Optimization for server SLOs is covered by SpecServe [16] and SmartSpec [32].

Edge-cloud collaborative LLM inference. The cascading pattern SKYCASCADE targets has been studied by SLED [28], SpecEdge [37], DSD [56], WISP [27], Galaxy [54], EdgeShard [57], SplitLLM [34], Jupiter [55], EdgeLLM [53], and Hybrid LLM [8]. Splitwise [39] and P/D-Device [19] address phase disaggregation (prefill vs decode) which is orthogonal to tier selection. None of these systems consider carbon.

Carbon-aware LLM and serverless scheduling. CarbonMin [6] routes per-call to low-CI regions. Sprout [25] shapes generation length under grid conditions. CarbonScaler [13] and CarbonFlex [14] provision cluster capacity for carbon efficiency. EcoServe [30] targets multi-hour serving traces. Caribou [12] performs geospatial shifting of serverless workflows. PCAPS [21, 22] introduces precedence-aware scheduling for data-processing DAGs. Wiesner *et al.* [49] characterize temporal shifting. Maji *et al.* [33] and Tabaeiaghdaei *et al.* [48] address multi-cloud and network routing. All operate at the call or workflow level and do not enter the cascading tier-selection decision.

LLM carbon characterization and modeling. LLMCO2 [11] predicts per-call carbon. Wilkins *et al.* [50] build offline energy-optimal models. Carbon in Motion [24] characterizes Open-Sora. Nguyen *et al.* [35] position the sustainable LLM serving problem. Li *et al.* [29] cover the LLM lifecycle. Hoffmann *et al.* [15] address federated LLM. Acun *et al.* [1] provide first-look carbon implications. Patel *et al.* [38] characterize cloud-LLM power management. Stojkovic *et al.* [45] survey LLM inference energy. Wu *et al.* [51] survey sustainable AI. The CO₂CoDe line [40] addresses HW/SW co-design.

Edge LLM sustainability. Husom *et al.* [17] profile quantized LLMs on Raspberry Pi. The Sustainability-Aware Edge work [41] measures Jetson Orin NX deployments. None of these works extends to multi-tier cloud cascading.

Agentic workload characterization. Pan *et al.* [36] provide the first large-scale empirical study of production agents (306 practitioners, 26 domains). OpenHands [2] and AgentDiet [52] measure token waste from failed plans and trajectory retries (18 to 50 percent reducible). Utilization fallacy work [43, 44] addresses GPU-utilization assumptions. Tokenomics [42] maps token consumption to activities. AgencyBench [26] provides agent workload benchmarks. DeepSeek-R1 [7] is the reasoning-model deployment that increasingly serves agent workloads. Sukprasert *et al.* [46] characterize the limits of carbon-aware shifting.

7 Discussion and Limitations

Privacy-pinned queries. Some mobile-LLM queries cannot leave the device or the regional edge for compliance reasons (medical, financial, jurisdiction-restricted). SKYCASCADE handles these by

allowing the eligible region set $R(P)$ to be empty at the cloud tier, in which case the cascade falls back to device-edge-only execution. The 17 percent of our workload that is privacy-pinned sees no SKYCASCADE benefit by construction; the 31.4 percent headline is on the remaining 83 percent. Production deployments with sub-1-percent latency overhead can set $\tau \in [0.5, 1.0]$ and recover 8 to 24 percent (Table 2, $\tau = 1.0$ row).

Marginal vs average CI. We use average carbon intensity throughout, the same choice Caribou makes [12]. Marginal CI would tighten the analysis [33], shifts to grids near renewable saturation may not displace fossil generation, putting our 31.4 percent on the upper end of what is achievable. The qualitative finding still holds, the cross-region variance is larger under marginal accounting (dirty-grid marginal higher than its average, clean-grid marginal lower), and the mechanism’s relative benefit grows. Only the y-axis magnitudes shift in Figure 4.

Embodied carbon and water. SKYCASCADE cuts operational carbon. The mechanism shifts per-region utilization with embodied-carbon implications (routing 30 percent of work to eu-north-1 instead of 1 percent amortizes its H100 fleet differently); the net effect is close to neutral in fleet-balanced deployments. Water also varies regionally [51], eu-north-1 is both carbon- and water-optimal but the alignment is not guaranteed, and a water-temperature parameter is a natural extension.

Scope. The addressable workload share is large in the field. Pan *et al.* [36] find 66 percent of production agents tolerate minute-scale latency and 15 of 20 case studies run asynchronously, the operating regime where $\tau \in [2.0, 4.0]$ recovers 31 to 39 percent at 3 to 7 percent overhead. Real-time agents (5/20, voice) stay at $\tau \in [0.5, 1.0]$; library default is $\tau = 1.5$. High-aggregate deployment pairs with a CarbonScaler-style elasticity layer [13] when low-CI regions saturate, and the mechanism carries over to Azure/GCP candidate sets unchanged.

8 Conclusion

The cloud tier of mobile-LLM cascades pays 86 percent of per-query energy and carries the widest cross-region carbon variance, yet existing cascading schedulers route latency-only and leave the cascade itself uncovered. SKYCASCADE closes this with a carbon-temperature parameter on the tier-selection cost function, cutting 31.4 percent at a 2.8 percent p95 penalty. The change is small. The mechanism holds as grids decarbonize, where a 3x average-CI improvement still leaves a 14 to 18 percent cut. The deployment cost is one parameter and no kernel changes, which keeps the abstraction portable across the cascading-inference stack.

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